

ECO101: Introduction to Microeconomics

Alternative Assessment, Summer 2024

Total marks: 30 (will be converted to 15%)

Instructions:

- *Submit your answers in clear A4 size papers which should be stapled properly. Loose papers will not be accepted.*
 - *Answer the following questions in the given order only. Make sure to number your answers clearly.*
 - *Only handwritten answers will be accepted. Do not submit typed answers.*
 - *You must write your full name, ID, and section number on the first page.*
 - *Any assignment submitted after the time mentioned will not be considered.*
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Answer all questions:

1. The weekly demand function for butter in the province of Quebec is given as:

$$Q_d = 20000 - 500P_x + 25M + 250P_y, \text{ where}$$

Q_d is the quantity in kilograms purchased per week,

P_x is the price of one kilogram of butter in dollars,

M is the average annual income of a Quebec consumer in thousands of dollars, and

P_y is the price of a kilogram of margarine in dollars.

Assume that $M = 20$, $P_y = \$2$, and the weekly supply function is such that the equilibrium price of one kilogram of butter is $P_x = \$14$.

- Find the quantity demanded of butter in kilograms. **[2 marks]**
- The price of a kilogram of margarine increased to $P_y = \$4$. In response to changes in the price of margarine, calculate and comment on the cross-price elasticity of the demand for butter given all other factors (P_x and M) remain the same. **[3 marks]**
- Say that the average annual income of consumers now increased to $M = 22$. Assuming P_y remains unchanged at $P_y = \$2$, calculate and comment on the income elasticity of demand for butter at the equilibrium. **[3 marks]**
- The price elasticity of demand for butter is -0.5 . What significance does this fact hold for suppliers of butter? **[2 marks]**

2. For each of the following scenarios, draw a demand-supply diagram to illustrate and explain the effects of the given shocks on the equilibrium price and quantity in the specified competitive market. Assume that each market is initially at equilibrium:

a. In the Banana Republic, more people are willing to buy imported cars as their income has increased. However, at the same time, the government imposed very high customs duty on imported vehicles to reduce traffic congestion and hence, importers are not bringing in as many cars as before. Draw an appropriate diagram and explain what will happen in the market for imported vehicles. **[5 marks]**

b. During the 2008 World Food Price crisis Bangladesh experienced a shortage of local rice production due to natural disasters. Simultaneously, obtaining necessary rice imports from other countries became challenging due to the global food price shock. This dual impact resulted in a severe rice supply crisis. Some studies also suggested that households suffering from income losses due to the natural disasters were forced to reduce rice consumption moderately. Assuming the reduction in consumption was less substantial than the impact of the artificial crisis, draw an appropriate diagram and explain how the equilibrium in Bangladesh's rice market changed. **[5 marks]**

3. The following table displays the production possibility of three cafes in an hour:

	Iced Coffee	Iced Tea
North End	3	5
Tabaq	5	6
Crimson Cup	2	4

a. What is **North End's** opportunity cost of producing 10 Iced coffees? **[2 marks]**

b. Which Cafe has an absolute advantage in making Iced coffees and Iced Teas? **[2 marks]**

c. Which Cafe should specialize in producing Iced coffee? Justify your answer using the concept of comparative advantage. **[3 marks]**

d. Which Cafe should specialize in producing Iced tea? Justify your answer using the concept of comparative advantage. **[3 marks]**